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"ANY COUNTRY, INCLUDING TRUSTED SOURCES, CAN INJECT A VULNERABILITY. WE SHOULD INSTAL A SYSTEM OF RIGOROUS VERIFICATION TO CHECK FOR WEAKNESSES IN COMPONENTS FROM ANYWHERE"

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The K3 Scout episode has a lesson for India, which is rapidly expanding its military UAV fleet while simultaneously trying to build an indigenous drone ecosystem. The next threat to a military UAV may not come from a missile, jammer or a counter-drone unit, it may already be embedded in one of its own components.

The K3 Scout scare exposes a growing weakness in modern defence supply chains: a platform can come from a trusted manufacturer and still contain a component whose antecedents are unknown and whose function has not been independently verified. For a military drone operating around sensitive bases, borders or operational zones, an unknown communication pathway can cause information leakage. Given that India-made platforms mostly depend on globally sourced cameras, processors, sensors, communications modules, semiconductors and software, the danger is clear.

According to industry estimates, tactical drones (which provide intelligence, surveillance and reconnaissance, or ISR capabilities) worth around Rs 3,000-4,000 crore have been delivered to the Indian armed forces since Operation Sindoor in May 2025, while they are expected to acquire additional tactical and armed UAVs worth at least Rs 15,000 crore over the next two years. Between them, the three defence services have 2,000-2,500 operational drones currently. The Indian Army alone is moving toward a goal of placing 8,000-10,000 drones per corps, aiming for an eventual 100,000-plus fleet size. The very scale of it all makes supply-chain security a national security issue.

Modern military drones are integrated systems wherein sensors, processors, navigation systems, flight controllers, communications equipment and software work as a single network. A vulnerability in any one can affect the security of the platform. Firmware—a microcode or software attached to the hardware System-on-Chip (SoC), which converts electrical pulses generated by camera sensors into digital data—and communications modules are particularly important. Firmware presents a special challenge because software can be altered remotely, without any physical change to the hardware, even after a platform has been certified. India's dependence on the global drone-component ecosystem becomes significant here, for China dominates the global drone components supply chain, creating dependencies in most sensitive components. And China is globally recognised as one of the main initiators and beneficiaries of electronic espionage.

However, the answer is not just to ban components from one country. "Any country, including your trusted sources, can inject a vulnerability," says Lt Gen. Karanbir Singh Brar (retd), distinguished strategic advisor at IIT Madras Pravartak, a technology innovation hub. "Hence, we should move beyond this limited understanding to instal a system of rigorous verification to check for vulnerabilities in components procured from anywhere." The K3 Scout scare, says Brar, demonstrates how a small component can create a dangerous weakness. "This is a serious signal for India," he adds.

SUPPLY CHAIN BLACK HOLES

In 2023, India restricted Chinese-origin components in military drones, particularly cameras, communications equipment, radio systems and software. Flagging potential security risks, procurement rules also placed restrictions on sub-components originating from countries sharing a land border with India. But mere country-of-origin screening cannot detect an undis-